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Unit 5 : Session 2 : SLO 1 | Parallelism Challenges Performance

I. Multiple Choice

1. b) Load balancing among processors.
2. c) The impact of the serial portion of a program — Amdahl's Law: $\text{speedup} \approx 1/(s + (1-s)/N)$; the serial fraction caps speedup.
3. b) Memory coherence issues — simultaneous access to shared memory creates inconsistent cached copies.
4. b) Synchronization overhead — grows with processor count and limits scalability.
5. c) Distributed-memory parallel systems — data must travel over an interconnect/network, so latency dominates.

II. True or False

1. TRUE — idle processors from load imbalance directly waste computational capacity.
2. FALSE — Amdahl's Law shows speedup saturates: the serial fraction bounds it regardless of processor count.
3. TRUE — multiple cached copies of shared data must be kept coherent, which is difficult.
4. FALSE — synchronization overhead increases as more processors must coordinate.
5. TRUE — communication delay is the dominant challenge in distributed-memory systems.

III. Scenario-Based Activity

- Challenge 1 — Load imbalance: some processors remain idle during computation phases, so the effective parallelism is far below 16; per Amdahl's Law the idle/serial phases dominate and cap the speedup (4 — instead of 16 —).
- Challenge 2 — Memory coherence/consistency: inconsistent output values show that processors read stale copies of shared variables; extra synchronization/coherence traffic needed to fix this adds overhead that further reduces speedup.
- Both explain the gap: uneven work wastes processors, and coherence traffic steals cycles from useful computation.